

(2.1) Pro nějakou sym. matici platí $\langle x, y \rangle = x^T A y$ a zároveň $P^T A P = I$, kde

$$P = \begin{pmatrix} 3 & 3 \\ 2 & 7 \end{pmatrix}, \text{ takže } A = (P^{-1})^T P^{-1}.$$

$$\text{Spočteme } P^{-1}: \left(\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{array} \right), \text{ tj. } P^{-1} = \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix}$$

$$\text{tak } A = (P^{-1})^T P^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 53 & -23 \\ -23 & 10 \end{pmatrix}.$$

$$\text{Takže } \langle x, y \rangle = x^T A y = (x_1, x_2) \begin{pmatrix} 53 & -23 \\ -23 & 10 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} =$$

$$= (53x_1 - 23x_2, -23x_1 + 10x_2) \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \underline{53x_1y_1 - 23x_2y_1 - 23x_1y_2 + 10x_2y_2}$$